PHYSICAL CUMULATIVE EXAM

APRIL 27, 1991

PART 1 of 2: At high temperatures gaseous iodine tends to dissociate and the vapor phase contains both I_2 molecules and I atoms. The extent of dissociation can be measured by monitoring the total pressure. Three sets of data were obtained by placing solid iodine in an evacuated vessel of volume 342.68 cm^3, and then heating to the desired temperature:

Temperature (Kelvin)	Mass of iodine (milligrams)	Total pressure (Pascals)
973	62.71	6327
1073	62.32	7599
1173	61.84	9303

From these data, find the equilibrium constant, K_p, and estimate the molar enthalpy of dissociation of I_2 at the mean temperature of 1073K.

$R = 8.314\ J\ K^{-1}\ mol^{-1}$

Atomic weight of Iodine: 126.9 g mol^{-1}

Standard pressure: 1×10^5 Pa